

NIR Spectroscopy Application for Quality Control of Wheat Grain Produced in Albania

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Abstract

Wheat and its processed flour-based items have been one of the main staple foods of Europeans, including Albanians. Its nutritional importance is attributed to its role as a crucial source of microelements and its versatility for conversion into many processed products, including bread, pasta, and pastries. Global marketplaces are concentrating on their quality indicators about safety and security. The study utilized near-infrared spectroscopy to assess the quality of domestically grown wheat by analyzing physicochemical characteristics, including protein, gluten, starch, moisture, and sedimentation index. This study included the application of NIRS in tandem with machine learning as a sustainable processing and quality assessment tool in cereals.

Seventy-five samples were gathered from various areas throughout the 2022 harvesting season, either straight from the field or from silos. The protein content varied from 9.6–15.0%, with a mean of 12.7%. The gluten concentration varied between 19.4% and 37.5%, with a mean of 27.5%. The starch content ranged from 66.3–71.5%, with a mean value of 68.9%. The sedimentation index was measured between 21.1 and 57.8 cm³, with an average of 37.6 cm³. The moisture level varied between 9.8% and 12.5%, indicating that all wheat samples remained within the maximum limit of 14.5% stipulated by national and European Union food regulations. The crop quality indicators generated in Albania met the quality criteria established by EU and state food legislation.

1. Introduction

Cereals are the primary type of seed food used by humans for millennia. They are considered staple foods in various geographical regions, including rice, wheat, and maize in this group. The *Triticum genus* includes wheat grain, which is part of the *Poaceae* family. Additionally, this family encompasses different cereals, including corn, barley, rice, and rye (Zhang et al., 2022). The Fertile Crescent was where wheat was first cultivated in ancient times, extending to Europe and Asia (Preece et al., 2017). *Triticum durum* and *T. aestivum* are the two wheat species, with the latter being commonly referred to as soft wheat, while *T. durum* is sometimes known as durum wheat. Soft wheat and its refined derivatives are indispensable in human nutrition. It is cultivated except in tropical climate regions (Wrigley, 2016). Grain seeds are typically ground, and flour is the commodity fraction used as food (Giraldo et al., 2019), but alternative consumption practices are also present.

Rice, wheat, and maize, known as three principal cereal grains, serve as staple foods and primary sources of carbohydrates in our daily diet. While consumption may vary across different geographical regions, wheat grain is the most important commodity for Europeans and the world population (Wrigley, 2016; Williams, 2006).

Wheat seed comprises bran, endosperm, and germ, each part having distinct chemical compositions. The seed is composed of carbs, proteins, and a smaller quantity of lipids. The endosperm is composed of significant amounts of proteins, carbohydrates, and iron; the germ provides vitamins, trace minerals,

and unsaturated and polyunsaturated fats. Usually, the bran is separated before consumption, even though it is abundant in dietary fibers (Javid Iqbal et al., 2022; Khalid et al., 2023).

Wheat stands out among cereals for its capacity to produce bread and dough products due to the gluten protein. Starch is the primary constituent of wheat flour, with proteins being the secondary component, which includes the gluten-forming gliadins and glutenin. The fat content is minimal in wheat seeds despite the species. It substantially contributes to our diet's energy, fiber, vitamins, and minerals. Its distinct characteristics give wheat special attention among other cereal grains. The proteins provide a cohesive texture to dough with the ability to stretch and bounce back, enabling the gas bubble formation and preservation, a mixture of air and carbon dioxide, during the bread-baking process (Wrigley, 2016; Javid Iqbal et al., 2022).

Agriculture, Albania's paramount sector, is not just a statistic but a vital part of the nation's economy and livelihood. It constitutes around 58% of employment and 21% of the nation's GDP (Dzartova-Petrovska et al., 2011). Although just 24% of the nation's overall area, Arable land is a crucial resource, with 43% situated in lowland regions with a comparatively high productive potential. Approximately 49% of the agricultural land is dedicated to cultivating feed crops (Diku, 2011).

Albanian agriculture's primary focus continues to be grain production, a tradition that is now evolving. The composition has transitioned towards providing animal feed rather than for human consumption. This shift reflects the changing needs and priorities in the agricultural sector. Livestock, now a significant part of the whole value of agricultural output, represents almost fifty percent. Crop and fruit production constitute 44% and 11%, respectively, although animal production represents over 50% of overall agricultural output (Gjeçi, Shytaj, & Biçoku, 2018).

Three agroecological regions are identified based on soil, climatic, geographical, and socio-economic characteristics: the Lowland region (0-200 m, above sea level, asl), a typical lowlands Mediterranean climate region; the Hill-land region (400–1000 m. asl) cultivates various crops and fruit trees; and the Mountainous region (1000–2700 m. asl) Wheat, barley, rye, and several other fruits are cultivated here (Diku, 2011).

In 2021, the global wheat output amounted to over 900 million tons, farmed in approximately 244 million hectares (FAOSTAT, 2022). Within the European Union, about 50% of the cultivated land dedicated to grain production is allocated to wheat. Albania predominantly cultivates maize and wheat as its primary grain crops, as seen in Table 1. Their cultivation is widespread nationwide in Fieri, Elbasan, Korça, and Tirana. The country's grain output for 2021 reveals that maize and wheat are the predominant crops cultivated, with wheat yielding a total of 225 thousand tons. The aggregate grain yield amounts to 691,353 tons of wheat, 225,171 tons of maize, 414,271 tons of barley, 15,310 tons of oats, and 33,309 tons of oats (INSTAT, 2022).

Table 1
Cereal production in Albania during the harvesting year 2021 (tons).

No.	District	Cereals	Wheat	Maize	Barley	Oats
1	Berati	32953	11636	17190	223	3904
2	Dibra	47266	7588	38516	302	227
3	Durrësi	41333	10404	29101	36	1792
4	Elbasani	98249	35297	57142	198	5341
5	Fieri	168386	62886	88517	4939	12044
6	Gjirokastër	16979	6327	8130	-	2522
7	Korça	85094	45950	26783	7983	2762
8	Kukësi	14586	2010	11271	-	547
9	Lezha	33138	11131	22007	-	-
10	Shkodër	52793	5105	47688	-	-
11	Tirana	46434	17558	25943	1409	1508
12	Vlora	54144	9280	41982	220	2662
Total		691353	225171	414271	15310	33309

Source: (INSTAT, 2022).

The expanding global population has intensified the pressure on the cereal markets, necessitating meticulous quality control on the cereal commodity supply chain. Concerns regarding the safety and quality of wheat flour are prevalent among the general public due to its direct association with the attributes of products derived from flour and human health (Zhang et al., 2022).

NIR (near-infrared spectroscopy) is a technique utilized to evaluate various characteristics of the quantity and quality of agricultural commodities (Baeten, 2019). The extensive application and worldwide recognition of moisture and protein content can be attributable to their substantial economic importance. The methodology's accuracy is associated with the near-infrared (NIR) model. Additionally, macroconstituents such as starch and fat, as well as essential micronutrients, including amino acids, dietary fibers, and amylose, are routinely analyzed by NIR spectroscopy. Furthermore, it is employed during processing to ensure quality control by evaluating flour moisture during milling or dough development. In addition, NIR is employed to grade and categorize substances according to characteristics, including vitreousness, color class, and the existence of insects. Finally, mycotoxins, contaminants, and ergot bodies in food are identified through their application (Johnson, 2020; Delwiche, 2021).

In contrast with conventional analytical methods, an extensive array of benefits consists of the NIRS application, including the cost reduction and time duration associated with the assessment procedure. This technique operates without prior sample preparation or chemical use, which confers notable benefits on an industrial scale regarding quality control and process monitoring (Baeten, 2019). NIR spectroscopy utilizes the 900–1700 nm spectral range to produce consistent and reliable information on food quality (Kamruzzaman et al., 2012; Munck, 2019). Its temporal range spans from the visible to the mid-infrared (Schuster, Huenb, Scherf, 2023). The complex structural information related to the chemical bond vibration C-H, O-H, and N-H is produced using the near-infrared region. These chemical groups are fundamental to organic molecules, such as sugars, lipids, and proteins, including water (Baeten, 2019; Delwiche, 2021).

The process of measurement consists of 1) spectral data obtained, 2) noise removal, 3) calibration model development through reference data, and (4) model assessment (Cen and He, 2007). Data analysis processing is essential in conjunction with fundamentals and instruments, constituting the first two pillars of NIR spectroscopy. The correlation between the distinct characteristics of the investigated sample and the transmittance or absorption values is established by analyzing data (Zhang et al., 2022).

Machine Learning (ML) is the scientific field in which machines are trained to learn without being strictly programmed (Samuel, 2000). It includes three main categories: (1) supervised learning (SL), (2) unsupervised learning (UL), and (3) reinforcement learning (RL). ML applications in agricultural production systems comprise different categories: (a) crop management, (b) livestock management, (c) water management, and (d) soil management. ML models have been applied in the multidisciplinary agri-technologies domain for crop management (61%), yield prediction (20%), and disease detection (22%), but they have never specifically accounted for mycotoxins' cooccurrence (Liakos et al., 2018).

NIRS is now one of the most promising modern analytical techniques due to its non-invasiveness, fast, non-destructive, high throughput, minimal/straightforward sample preparation, chemical-free, instrument portability, and suitability for use by non-specialist operators. The technology has a strong perspective for further development due to the advancement in optics, computing power, and the recent wave in data science, AI, machine learning, and, more recently, deep learning. It allows several constituents to be measured simultaneously from a single spectrum. Therefore, this project aims to use NIRS in tandem with machine learning as a sustainable processing and quality assessment tool for mycotoxin assessment in cereals.

2. Methods

2.1. Wheat samples

75 wheat samples were collected throughout the summer of 2022 from the central producing districts of Fieri, Elbasan, and Korça immediately after the harvest. The wheat samples were collected according to the sampling protocol and then stored in bags. Ultimately, they were transported to the laboratory and

placed in a refrigerated chamber at a temperature of 4°C in dark space until they were subjected to a physicochemical examination.

2.2. Physicochemical analysis

The standard method for measuring grain protein content often utilizes near-infrared (NIR) spectroscopy, which is applied to whole, milled, or flour. This method relies on the connection between NIR spectroscopy and total nitrogen studies. Approximately 75 representative samples of the entire material were subjected to laboratory testing. The provided samples were utilized to calibrate a partial least squares regression (PLSR) model. This model was subsequently employed to predict the quantities of water, protein, starch, β -glucan, fat, and moisture.

Per the specifications outlined in Standard EN 15948:2015, the InfratecTM 1241 employs a near-infrared wavelength region to analyze the kernel. This enables the examination of various parameters, including fat, hydration, protein, gluten, sediment, and starch in whole kernels. The apparatus incorporates the sample chamber, the light source monochromator, the wavelength selection processor, and the detector. A minimum bulk sample of 500 grams in weight is deposited into a hopper apparatus. The crucial data is generated via the reflectance mode within the 1100–2500 nm wavelength range. Under the greater quantity of light energy in the lower range, deeper penetration into the interior of the seed kernel is achievable (EN 15948:2015).

Based on Beer's law, the methodologies utilized to examine optical data about absorbed samples can be characterized as various approaches to establish a correlation between analyte concentration and sample absorbance (A) at a specific wavelength.

For calibration, the multi-regression equation is utilized as follows:

$$Y = Bo + Bi(-\log Ri)N + E$$

1

The variable Y represents the percent concentration of the absorber; Bo equals the intercept of regression; (Bi) Regression coefficient; i = index of the utilized wavelength and reflectance (Ri) of that wavelength; N = number of wavelengths utilized in total for regression; E = random Error.

3. Results

Being a rapid and non-destructive technique, near-infrared spectroscopy (NIR) has become increasingly popular in food and feed quality evaluation (Kamruzzaman et al., 2012; Zhang et al., 2022). It provides the decision-makers with a helpful acting tool to protect the quality of the grain and the products associated with it in storage and processing technologies. Seventy-five wheat samples from the harvesting season of 2022 were analyzed using FOSS NIR equipment, InfratecTM 1241. The analyzed

quality parameters included the protein content, moisture, starch content, gluten, and sedimentation index (Table 2).

Table 2
Chemico-physical parameters in wheat during the 2022
harvesting season (%).

Parameter	Min	Max	Mean
Proteins (%)	9.20	15.10	12.54 ± 1.27
Humidity (%)	9.50	12.20	10.86 ± 0.53
Starch (%)	66.80	72.00	69.59 ± 1.21
Gluten (%)	18.20	39.20	28.17 ± 4.34
Sedimentation index (cm ³)	19.20	59.00	39.35 ± 8.62

The range of protein content was from 9.2 to 15.1%, with an average of 12.5% (Fig. 2).

The near-infrared (NIR) examination of the starch in the wheat sample yielded a range of 66.8–72.0%, with an average value of 69.6% (Fig. 3).

The gluten concentration ranged from 18.2 to 39.2%, with 28.2% being the average number of gluten occurrences.

It turns out that the sedimentation index ranged from 19.2 to 59.0 cm³, with an average value of 39.35 cm³ (Fig. 5).

Optimal moisture levels are crucial for the proper storage of flour. Elevated flour moisture levels facilitate mold and worms' proliferation throughout storage. As per EU law, the moisture content ranged from 9.5–12.2% in the wheat samples, and none of the samples exceeded the maximum threshold level of 14.5%. The mean protein content values were 12.54 ± 1.27%, with a high of 15.1%.

4. Discussions

This work aimed to apply NIR spectroscopy as a rapid technique for analyzing the physicochemical characteristics, including protein, gluten, starch, sedimentation index, and moisture, in wheat grain cultivated and consumed in Albania. Near-infrared spectroscopy (NIRS) has long been utilized in the food and agricultural industries to assess the protein levels in grains, particularly wheat accurately. NIRS application in wheat quality control serves as an analytical tool to determine grain composition precisely and, secondly, applied as a fast, non-destructive technique during seed screening in the breeding process (Delwiche, 2021; Williams, 2008). The successful measurement of wheat protein content is attributed to the robust and broad absorption of the N-H bonds in the NIR spectral region (Caporaso, Whitworth & Fisk, 2018). The NIR monochromator with a 400–2500 nm range was used to scan entire grain samples

for reflectance analysis. The Partial Least Squares (PLS) technique is applied to establish calibration equations for the qualitative attributes of whole wheat (Pojić & Mastilović, 2013). The advantage is the time and cost of completing the test in a few minutes. Compared to conventional laboratory methods, the cost-effective tests lack solvents and other expensive laboratory equipment. This is particularly important in industrial-scale quality control (Cen and He, 2007; Kahrman and Egesel, 2016).

The calibration model establishes a correlation between the spectral data and the molecule (or property) of interest. The formulation of a NIRS calibration model proves challenging due to the complex characteristics of the samples under examination. Consequently, numerous interference bands are generated, in which the characteristics or components of relevance tend to coincide. NIRS devices, chemometrics approaches, and computer technology have significantly advanced the NIRS methodology (Williams, 2008).

Gluten content is a key factor for evaluating the quality of wheat flour, as it correlates significantly with baking quality. Protein content, which can vary within a range of 8–16%, alone does not consistently serve as a dependable indicator of baking quality, as it has been observed that flours featuring comparable protein content but distinct gluten compositions manifest divergent baking characteristics (Schuster et al., 2023). Protein and glucose constitute the principal components of wheat flour, comprising the intervals of 10–12% and 70–75%, respectively. Carbohydrates and lipids, on the other hand, are minor constituents, accounting for approximately 2% each (Goesaert et al., 2005). During the process of dough mixing, proteins undergo conformational changes, resulting in a three-dimensional structure that contributes to the strength and chewy texture of the final baked items.

The protein composition of several commonly consumed cereals exhibits variation. For instance, barley has around 9–12% of the total weight of the whole grain, maize contains 8–12%, oats include 12–15%, rye contains 12–15%, and wheat contains 9–16%. The protein level of this wheat variety is more significant (12–16%) compared to soft wheat, particularly durum wheat (Williams, 2006; Bekes and Wrigley, 2016). The examination of protein content is assessed by several techniques (Table 3).

Table 3
Methods applied to determine the protein content of cereal grain wheat.

Process	Material assayed	Method
Mill	Grain/ Flour	NIR
Extract	Protein in solution	Biuret
Hydrolyze	Amino acids	Liquid chromatography
Alkaline digestion	Amide N	Titrate ammonia released
Acid digestion	Ammonia or nitrogen	Kjeldahl method

Source: Bekes and Wrigley (2016).

Starch is the predominant macromolecule found in plants, consisting of helical chains of glucose molecules joined together by α -1,4 bonds and connected by α -1,6 bonds. It is present in the form of granules within the grain's endosperm. Starch consists of two primary components: amylose, composed of linear chains of α -(1–4)-linked glucose, and amylopectin, which is often the predominant component and consists of extensively branched polymers (Wrigley, 2016). The near-infrared (NIR) measurement of the starch in the examined wheat revealed a range of 66.8–72.0%, with an average of 69.6%.

Plant proteins account for almost 50% of the protein in our diet. The protein harvest comprises wheat, rice, and maize, the three primary cereal crops (Bekes and Wrigley, 2016). The gluten concentration ranged from 18.2–39.2%, with an average of 28.2%. The recommended threshold for the gluten content in wheat flour is around 24% when measured in its wet state (Kaushik et al., 2015). Out of all the wheat samples studied, just four, which make up 5.3% of the total, had a gluten concentration below 24%. This suggests that the wheat produced in 2022 is of exceptional quality.

The sedimentation index was measured in the interval 19.2 and 59.0 cm³, with an average value of 39.35 cm³. This test is a scientific examination that offers valuable information about the baking properties of wheat flour. The Zeleny value measures the sedimentation level of flour suspended in a lactic acid solution during a certain period. The Zeleny test value is employed to evaluate the quality of the baking process. Higher gluten content and enhanced gluten quality lead to lower sedimentation rates and higher Zeleny test results (Hruskova and Famera, 2003).

Optimal moisture levels are crucial for the proper storage of flour. Elevated flour moisture levels facilitate mold and worms' proliferation throughout storage. As per EU legislation, the moisture content ranged from 9.5–12.2% in the wheat samples, and none exceeded the maximum threshold of 14.5%. The mean protein content values were $12.54 \pm 1.27\%$, with a high of 15.1%. An important global issue in food safety is the presence of mycotoxins in grains. The presence of both controlled and non-regulated mycotoxins poses an ongoing risk to the global population, primarily because grains such as wheat and maize are essential staples in many diets. Prolonged exposure can result in various toxic detrimental effects on specific organs. The problem of mycotoxin contamination in Albania has received attention, particularly

concerning the contamination of cereals (Topi et al., 2017; Topi et al., 2023). However, NIR practitioners have focused on other components, such as aflatoxin in corn or deoxynivalenol in wheat, even though their natural concentrations in large quantities are below the detection limits of NIR reflection or transmission spectroscopy at levels lower than one part per million (sub-ppm. (Levasseur-Garcia, 2018; Mato et al., 2024).

5. Conclusions

The NIRS application analyzes quality indicators such as protein, gluten, starch, and sedimentation index in whole-kernel wheat grown in Albania. Ensuring the quality of wheat and wheat flour is of significant economic importance in food safety and security. The utilization of NIR spectroscopy for cereal quality study demonstrates that wheat commodity production in Albania exhibits exceptional quality. Climate change affects both the ability to produce enough food and the safety of grain production. An investigation of the physicochemical properties of wheat grain harvested in 2022, such as protein, gluten, starch, sedimentation index, and moisture, found that the protein content varied between 9.2% and 15.1%, with an average of 12.5%. Four out of the evaluated samples fell below the threshold of 10.5%. Using affordable and easily transportable methods like Near-Infrared Spectroscopy (NIRS), the chain production from farms to the mill may enhance grain output and raise awareness about the economic significance of quality metrics.

Abbreviations

ASTM

American Society for Testing and Materials

EU

European Union

EN

European Norm

FAOSTAT

Food Agriculture Organization Statistics

INSTAT

Institute of Statistics of Albania

NIR

Near Infra-Red

NIRS

Near Infra-Red Spectroscopy

PLS

Partial least squares

PLSR

Partial Least Square Regression

Declarations

Ethics approval and consent to participate

The wheat grain used in the study was collected from various farmers of the main producing agricultural regions of Albania. They were collected and appropriately documented during the exploration program duly permitted and approved by the university under the national research project financed by the National Agency of Scientific Research and Innovation. The collection of the grains used in the study complies with local or national guidelines, with no need for further affirmation. As a member of the University and Head of the Laboratory of Toxic Substances and Biomolecules, the corresponding author (DT) is fully authorized to collect, conserve, and characterize the local crop resources for quality control and contaminant analysis. All the guidelines were followed per the university's research ethics for collection, characterization, and documentation. Any opinions, findings, conclusions, or recommendations expressed in this publication are those of the author(s) and do not necessarily reflect the view of the University of Tirana or NASRI.

Consent for publication

The authors have approved the consent for publication in advance.

Competing interests

The authors declare that there is no disclosure of potential conflicts of interest.

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Author Contribution

A.T. writing manuscript, data curation L.M. writing manuscript, sampling, A.O. analysis, validation, reviewing D.T. funding, monitoring, reviewing N.B. writing manuscript, sampling

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Data Availability

Data is provided within the manuscript or supplementary information files.

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Figures



Figure 1

FOSS infrared device (Infratec™ 1241).

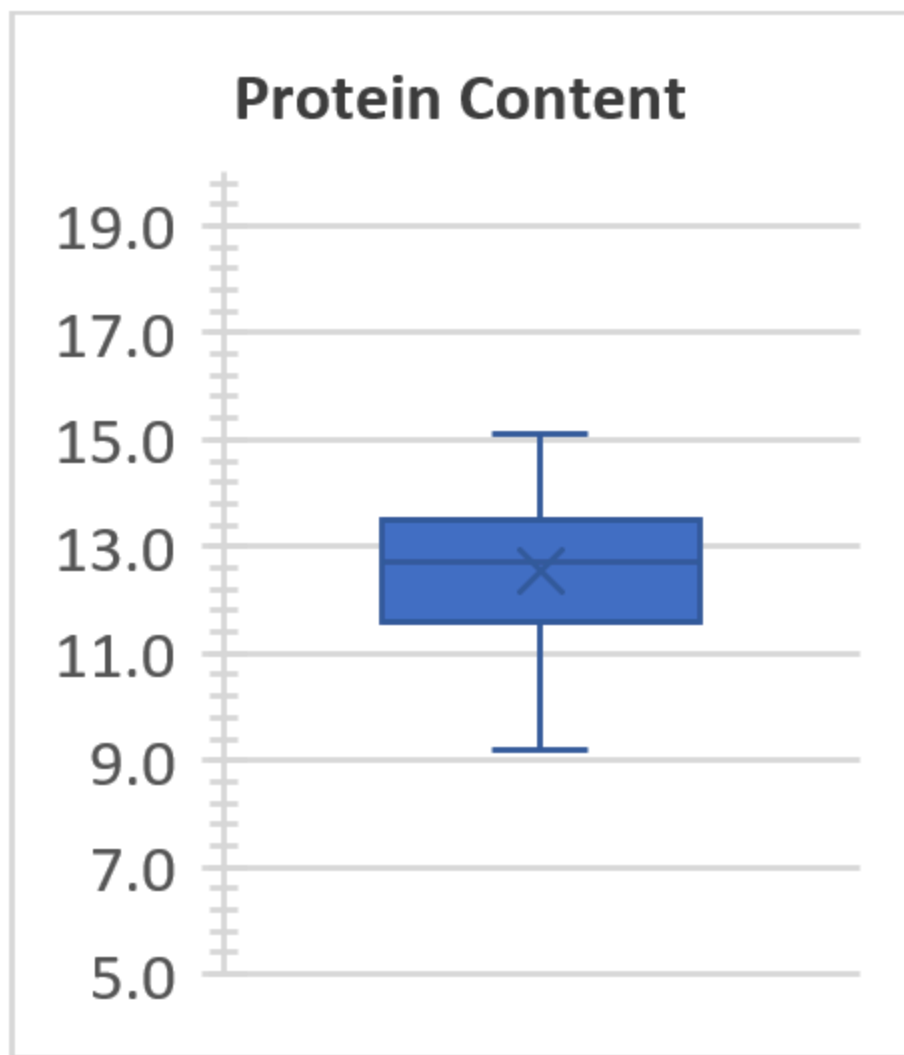


Figure 2

Protein content in wheat samples (%).

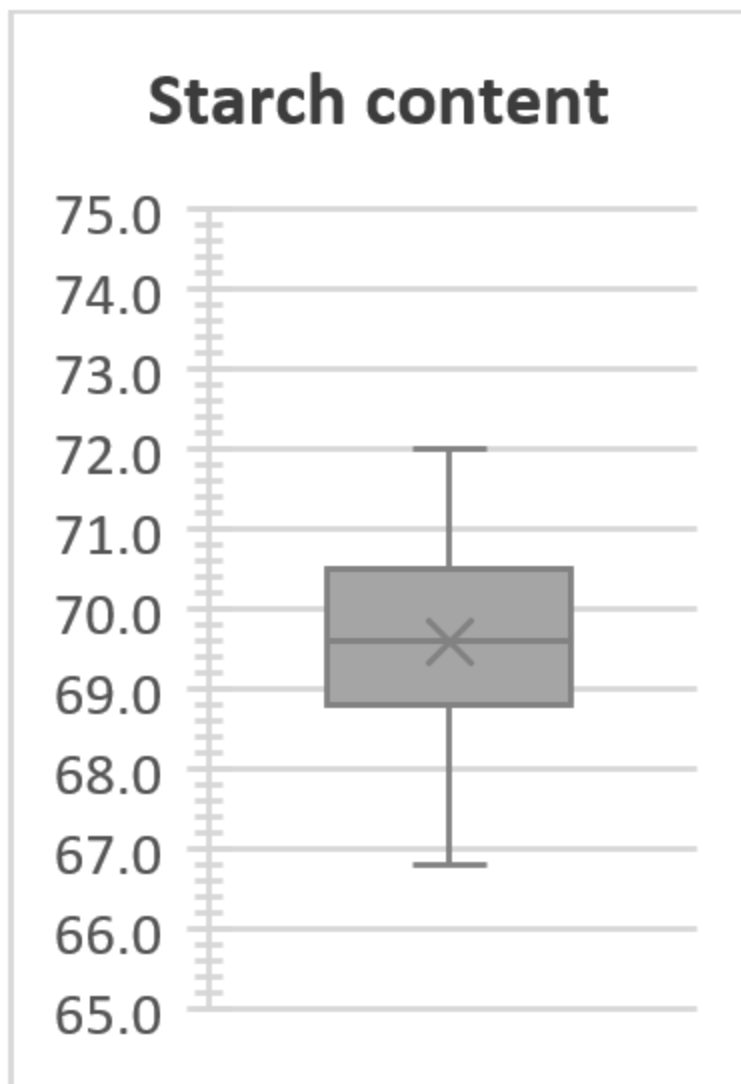


Figure 3

Starch content in wheat samples (%).

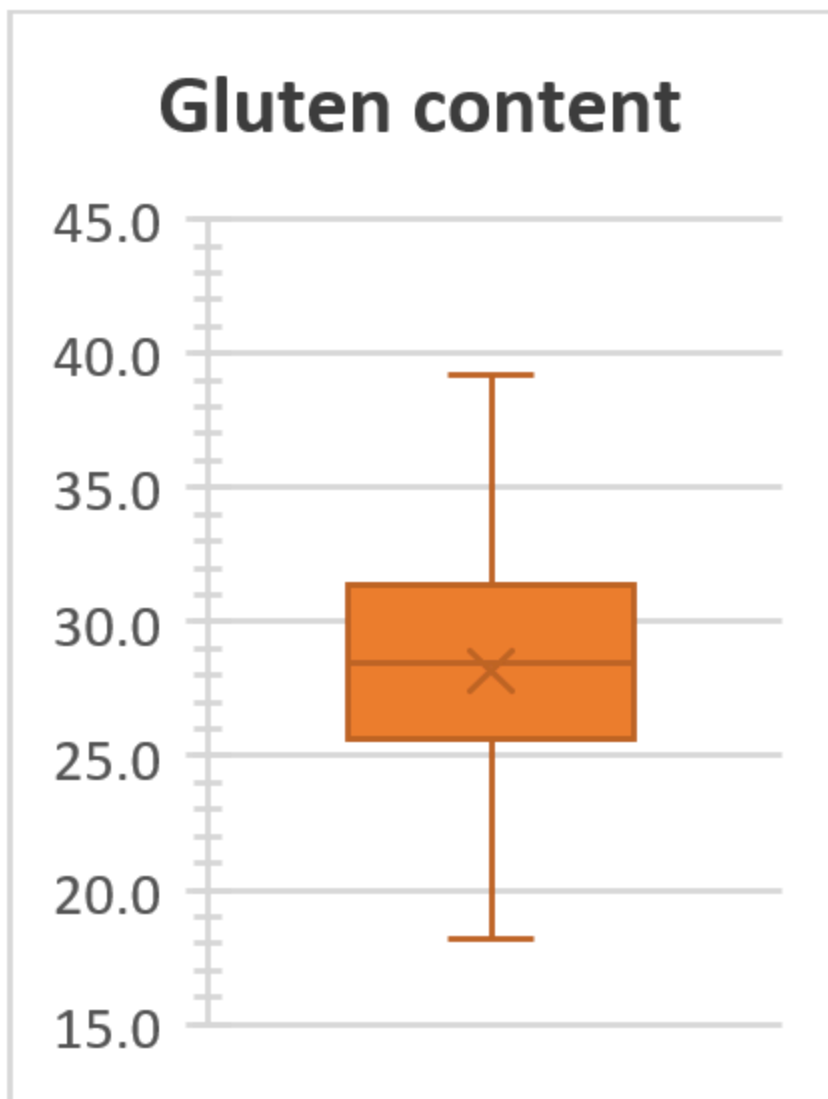


Figure 4

Gluten content in wheat samples (%).

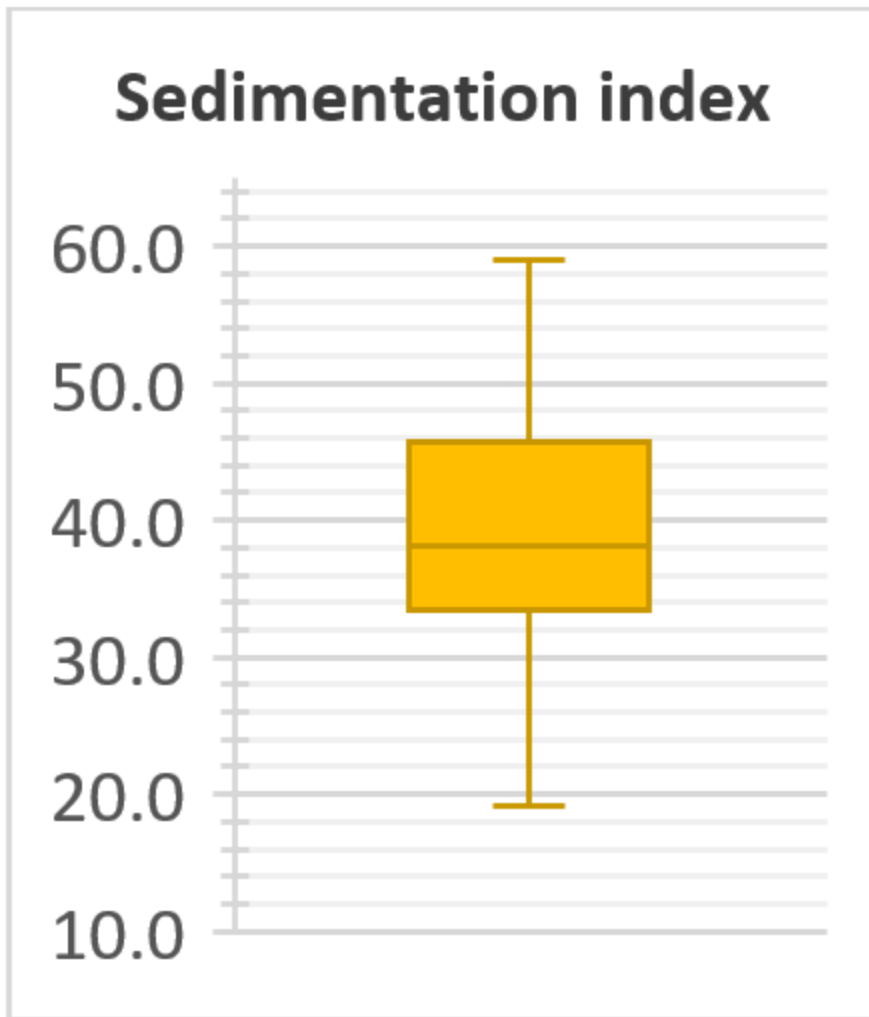


Figure 5

sedimentation index (cm³).